

CAMPBREAK · PRECIS

CampBreak design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/campbreak		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-03	Amish Chadha	Amish's round-2 decision 1A (CBK-DDR-004): delivery hose stowed full; cart mass, pull and R9 figures updated
0.4	2026-10-03	Amish Chadha	Amish's R9 decision 1A (CBK-DDR-003): check foot valve, suction hose left coupled, go when two arrive; figures updated
0.3	2026-10-03	Amish Chadha	TRL 3 with the decisions of CBK-DDR-001 and CBK-DDR-002 applied and results from CBK-CAL-001
0.2	2026-10-03	Amish Chadha	Populated to TRL 2 with concept media, components and first-order numbers
0.1	2026-09-30	Amish Chadha	Initial scaffold

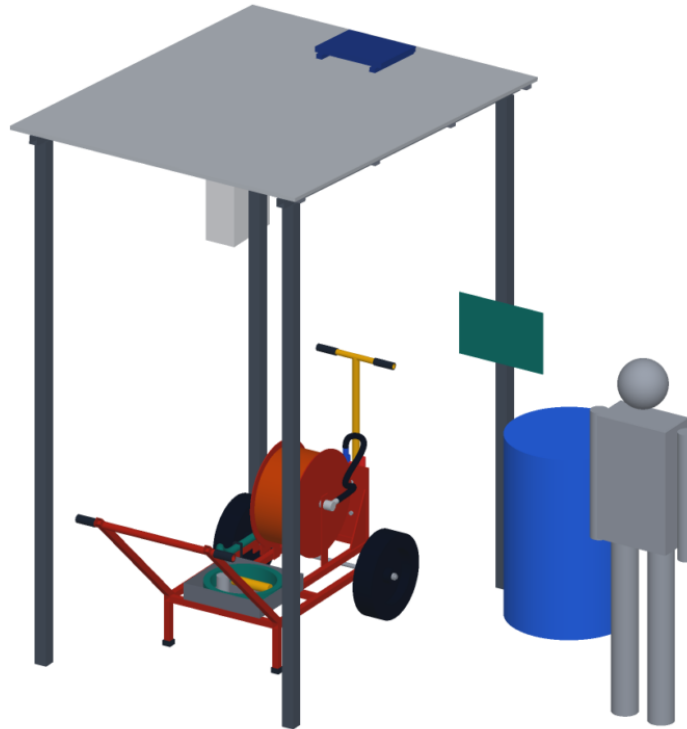
CampBreak design precis

Gives camp residents block alarms and a hand-pumped hose cart to fight shelter fires in the first minutes.

CampBreak

github.com/BoujeeEnjinia1701/campbreak

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Hose cart parked in its block station beside the 200 L drum. Grey figure: 1.75 m person for scale

Figure 1. Concept render of the block station and hose cart, from the TRL 3 model.

■ How it works

Each shelter in a block gets a battery heat alarm hung under a roof pole. When the air at the alarm warms by 8 °C per minute or more, or passes 57 °C, the alarm sounds and radios the block station. The station sounds a 110 dB(A) siren and broadcasts once, so every alarm in the block sounds within about 7 s. Volunteers run to the station, pull out the two-wheeled hose cart and take it to the burning shelter. There they drop the suction hose into a water drum, jerrycan or open container (or connect the tap adaptor to a standpipe), run out the hose from its reel, and two people pump the T-bar lever to drive about 20 L/min through a 6 mm jet that reaches about 7 m. The aim is to keep a fire to one shelter while others clear the neighbours and call the fire service.

CampBreak: water pressure from the hand pump to the nozzle at 20.4 L/min, two people pumping (all values are estimates; relief valve set at 400 kPa)

github.com/BoujeeEnjinia1701/campbreak

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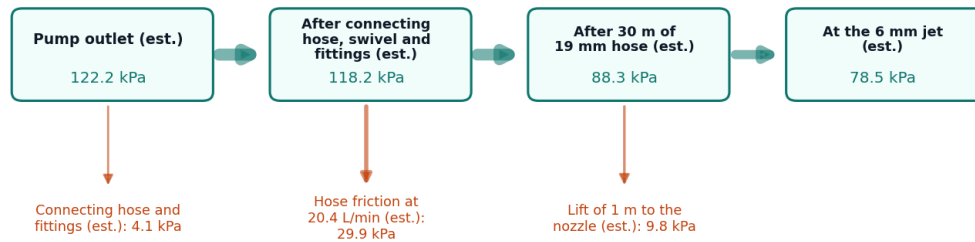


Figure 2. Pressure along the water line at 20.4 L/min with two people pumping; all values are estimates from CBK-CAL-001.

■ Components

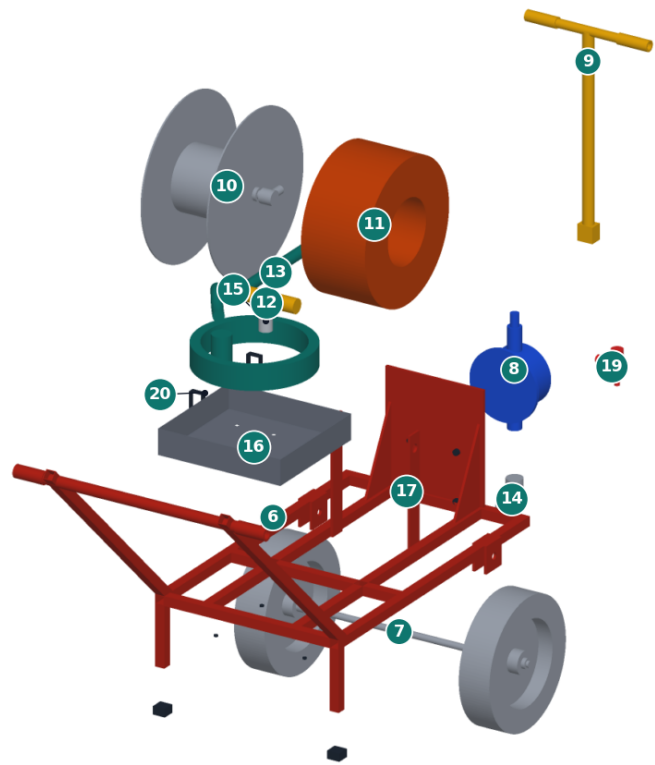
Table 1. Main components (numbers match bom/bom.csv).

BOM	Component	What it is
1	Heat alarm (24 per block)	Stock ABS box 100 x 100 x 40 mm with a controller and sub-GHz radio board, a thermistor below the box in a printed guard, a 95 dB(A) piezo and three AA alkaline cells; hung under a roof pole on an aluminium plate and two cable ties
2	Station alarm box, siren and solar panel	Locked IP65 box with the receiver board, charge controller and 12 V 7 Ah sealed lead-acid battery; 110 dB(A) siren; 10 W panel on the roof
3	Station frame and roof	Four 60 mm square posts in concrete collars, two beams, four purlins, corrugated sheet; about 1.5 x 2.05 m of shade
4, 5	Sign and water drum	Instructions in the camp languages; a 200 L drum beside the cart
6	Cart frame	Welded 30 mm square tube frame with pull handle, pump stand, reel uprights and axle plates
7	Wheels	Two 400 mm puncture-proof wheels on a 20 mm through axle
8, 9	Hand pump and lever	Cast-iron double-acting semi-rotary pump, 25 mm ports, with a T-bar lever for two people
10, 11	Reel and hose	30 m of 19 mm semi-rigid hose on a reel with a swivel inlet, fed from the pump by a short connecting hose
12 to 15, 20	Nozzle, suction hose, couplings, tap adaptor, hose straps	Jet and spray nozzle with shut-off; 4 m of 25 mm suction hose with a brass check foot valve, left coupled to the pump inlet and strapped along the left rail so the pump stays primed; cam-lever couplings; a push-on tap adaptor
16	Hose tray	Folded galvanised tray on the cart that carries the suction hose and small parts
19	Relief valve	4 bar spring relief valve at the pump outlet

CampBreak: hose cart, exploded viewgithub.com/BoujeeEnjinia1701/campbreak

CONCEPT, NOT FOR FABRICATION

- 6 Cart frame, welded
- 7 Wheels, axle, spacers and collars
- 8 Hand pump, semi-rotary, 25 mm ports
- 9 Lever extension with T-grip
- 10 Hose reel drum, spindle and swivel inlet
- 11 Delivery hose 19 mm x 30 m and connecting hose
- 12 Nozzle, jet and spray, with shut-off
- 13 Suction hose 25 mm x 4 m with check foot valve
- 14 Quick couplings, 25 mm
- 15 Tap adaptor
- 16 Hose tray
- 17 Fasteners
- 19 Pressure relief valve, 4 bar
- 20 Suction hose straps

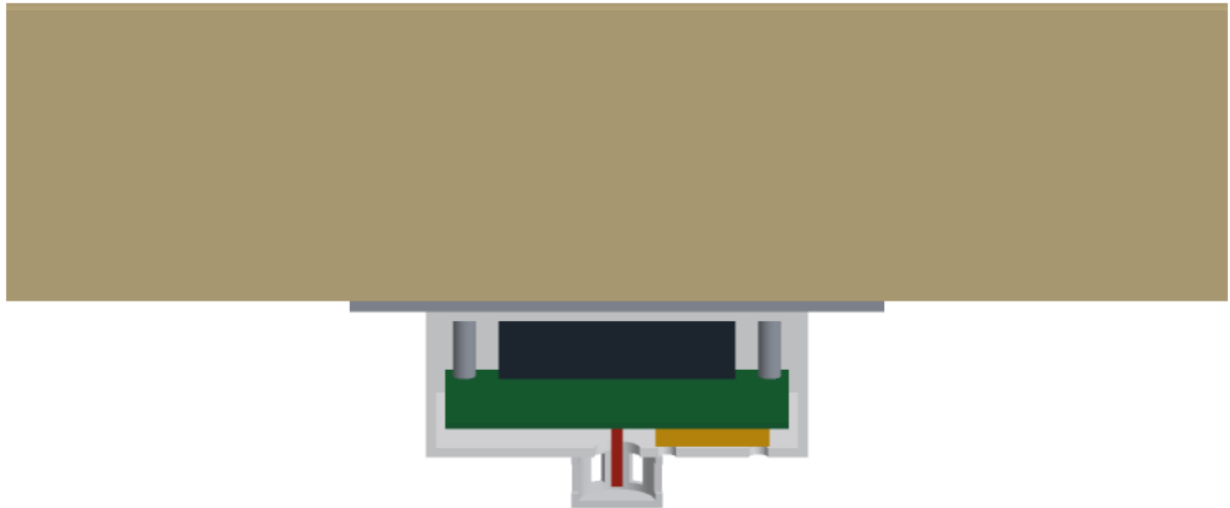


Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 3. Exploded view of the hose cart; callouts match bom/bom.csv.

CampBreak: heat alarm cutawaygithub.com/BoujeeEnjinia1701/campbreak

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation. The alarm hangs under a roof pole; box 100 x 100 x 40 mm

Figure 4. The heat alarm cut open: battery holder, board, sounder and the thermistor below the box.

■ Key design choices

All decided under Amish's 2026-10-03 pre-approval (CBK-DDR-001 and CBK-DDR-002):

1. **Heat, not smoke.** Rate of rise of 8 °C per minute with a 57 °C fixed back-up, because shelters cook with open flames.
2. **Star relay, published in full.** Alarms report to the station, which broadcasts once; no mesh.
3. **A bought pump.** A semi-rotary wing pump is sold in most target markets and is repaired with hand tools.
4. **No lithium cells.** Alkaline cells in the alarms and a sealed lead-acid battery at the station.
5. **Relief valve at 4 bar** so a shut nozzle cannot over-pressure the hose or fittings.
6. **A cart two people can pull,** 850 mm wide, about 103 kg loaded and primed, with the delivery hose full, on 400 mm puncture-proof wheels.
7. **Stations sited within 70 m of every shelter** as the siting margin for the 3 minute target.

Decided by Amish on 2026-10-03 ("1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"; CBK-DDR-003):

1. **A primed pump and a coupled suction hose.** A check foot valve keeps the pump and suction hose full, and the hose stays coupled to the pump, so nothing is coupled or primed at the fire.

2. **Go when two arrive.** The first two volunteers at the station take the cart, within 30 s of the siren; the rule is drilled and printed on the drill card.

■ First-order numbers

Table 2. Key figures (CBK-CAL-001; estimates unless stated).

Quantity	Value	Main assumption
Flow at the nozzle	20.4 L/min	0.40 L per double stroke at 60 per minute, 85 % volumetric efficiency
Jet reach	6.9 m	Half the drag-free range of a 12.5 m/s jet at 30°
Effort per person	47 W, about 28 N at the grip	50 % mechanical efficiency
Pump outlet pressure	1.2 bar	30 m of 19 mm hose, nozzle 1 m above the pump
One 200 L drum	about 10 min	
Alarm sound at 3 m	85.5 dB(A)	95 dB(A) at 1 m
Alarm relay, worst case	7.2 s	Three retries, station broadcast heard within 4 s
Alarm battery life	Cell shelf life (about 5 years)	6.9 μ A average
Cart mass, loaded and primed	103 kg	Catalogue masses for bought parts; 2.6 kg of water in the pump and suction hose and 8.9 kg in the delivery hose
Pull on a 10 % slope	101 N per person	Rolling resistance 0.10; 1 N over the 100 N assumed sustainable
Water on target	2.95 min at 100 m; 2.5 min at 70 m	1.0 m/s walking, 30 s gather, delivery hose stowed full
Block kit cost	USD 1,374	bom/bom.csv

Value-engineering target: USD 1,600. Estimated cost of the constructable design: USD 1,374 (USD 226 under the target).

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- Rate-of-rise heat sensing, as the preliminary screen advised; no alarm certification is claimed.
- Lumkani's networked heat detector has no patent identified in the preliminary screen, but its mesh network is described as patented. CampBreak uses a plain star scheme (alarm to station, station broadcast to all), published in full, and is checked again before release.

■ Shared blocks

- FieldNode sensor core (power and radio lessons for the alarm and block station)

- CalRig proof-load (pressure and hose proof testing)
- BreakHook (sibling: opening firebreaks in dense settlements)

■ Safety

Safety: The kit is for small, early fires only. People leave first; nobody enters a burning shelter or stays when the fire spreads, and the fire service is called at once. Never use water on burning cooking oil or on live electrical wiring.

Safety: The pump, hose and nozzle are pressurised. The relief valve opens at 4 bar and points its discharge at the ground; the hose is rated at 10 bar. Check the hose, clips and couplings before every drill, never point the jet at people, and shut the nozzle only while someone is still at the pump.

Safety: The station battery is sealed lead-acid: fuse it at the terminal, keep it in the locked box, and recycle it through a battery dealer. The alarms use alkaline AA cells; follow the camp's disposal rules. There are no lithium cells in the kit.

Safety: The loaded cart weighs about 103 kg, with the delivery hose full. Two people pull it; on slopes steeper than 10 % a third person holds it back. Park it on its feet with the handle down.

This design is published as an open engineering reference by Design Molecule Labs. It is not certified equipment.

■ Open questions

None at design level. Items that can only be settled with real parts are listed in the design decisions register ([docs/06-design-decisions.md](#)).